

Does Hybridisation Pay? A Systematic Review of Metaheuristic Optimisation and Machine Learning Hybrids in Pavement Engineering (2005–2026)

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Abstract

PLACEHOLDER — structured abstract, ~280 words, written last so that every number in it traces to the audit. Do not draft this before Section 9 is final.

Keywords: pavement engineering, metaheuristic optimisation, machine learning, hybrid models, systematic review, methodological rigour

Draft status. This file renders to PDF, Word and HTML from one source. Sections marked *PLACEHOLDER* are not yet written. Every citation resolves against `references.bib`, which is generated from `data/seed_bibliography.csv`; a citation key that is not in the verified database fails the render rather than passing silently.

1. Introduction

Pavement engineering has, over the past decade, adopted a specific and recurring rhetorical move: an optimisation algorithm is coupled to a learning model, the coupled system is shown to out-predict some comparator, and the coupling itself is offered as the paper’s contribution. Grey wolf optimisers, whale optimisation, teaching-learning-based optimisation, forensic-based investigation, dipper-throated optimisation and dozens of similarly named metaheuristics have each, in turn, been married to a neural network, a support vector machine, an extreme learning machine or a gradient-boosted ensemble, and the resulting hybrid reported as an advance over the base learner alone. The pattern is now common enough that “hybrid” is arguably the single most frequent claim of novelty in the data-driven pavement literature of the 2020s.

This review asks a question that the pattern itself rarely answers: **relative to what, exactly, is the hybrid better?** A hybrid model is compared, almost without exception, against a base learner running with default or lightly-adjusted settings. It is compared far more rarely against the same base learner tuned

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by a conventional procedure under a comparable search budget — the baseline that would isolate what the metaheuristic actually contributed. Where that comparison has been made, in the small number of cases we have located so far, the hybrid’s advantage over a properly tuned baseline shrinks, and sometimes reverses. Zeiada et al. [1] individually tuned eight learners, including deep architectures, against the Witczak and Hirsch mechanistic models for asphalt dynamic modulus; the winner was a tuned bagging ensemble, not a hybrid and not a deep network. Azam et al. [2] compared six swarm-optimised variants of one base learner against each other and report top-three results within a fraction of a percentage point of R^2 — a separation well inside what a different random seed would produce — but never against a conventionally tuned version of the same learner, so no premium can be read from the paper at all. These two studies, sitting side by side in the same sub-field, illustrate the gap this review sets out to measure at scale: **how often is the hybridisation premium — the gain attributable to the coupling itself, over a fairly tuned baseline — actually reported, and how large is it when it is?**

Answering this requires solving a prior problem, which turns out to be nontrivial in its own right: **finding the studies**. A literature search built on the word “hybrid” misses a substantial share of the relevant work, because many qualifying studies describe themselves with the vocabulary of optimisation rather than of hybridity — “optimised”, “improved”, “novel X–Y” or simply the two algorithm names side by side. Alhussan et al.’s dipper-throated-optimisation random forest for pothole classification is a case in point: widely cited, structurally a textbook metaheuristic-on-learner hybrid, and retrievable by a search on algorithm names but not reliably by a search on the word “hybrid” itself. Section 2 describes the algorithm-driven search design this observation motivates, and reports, as one of the review’s descriptive findings, what share of the eligible literature would have been missed by a label-driven search alone.

1.1. Positioning relative to the existing review literature

Prior reviews of artificial-intelligence methods in pavement engineering are numerous but occupy different territory, summarised in Figure 1. The most direct predecessor, Yang et al. [3], catalogues where neural networks have been applied across the pavement lifecycle; it stops at the MLP/CNN/RNN generation and does not treat hybridisation, optimisation coupling or methodological rigour as organising questions. Domain-specific vision surveys [4, 5, 6] cover distress detection in depth but do not extend to materials, structural evaluation or network-level decisions. Two very recent contributions come closer to the diagnosis this review develops: Tamagusko and Ferreira name standardisation, interpretability and replicability as open problems specifically for roughness prediction on flexible pavements, and M’harzi Alaoui et al. [7] describe an “interpretability paradox” and a lab-to-field gap across 180 soil-stabilisation studies. Both are right, and both are confined to a sub-domain and stop short of a per-study coding instrument. What has not been done, in any of this literature, is to (i) define hybridity structurally rather than lexically so that the algorithm- vocabulary studies are not missed, (ii) code a consistent rigour rubric across the full range of

pavement sub-domains rather than one at a time, and (iii) convert the resulting evidence into a quantified premium and a reusable reporting instrument. That is the gap this review occupies.

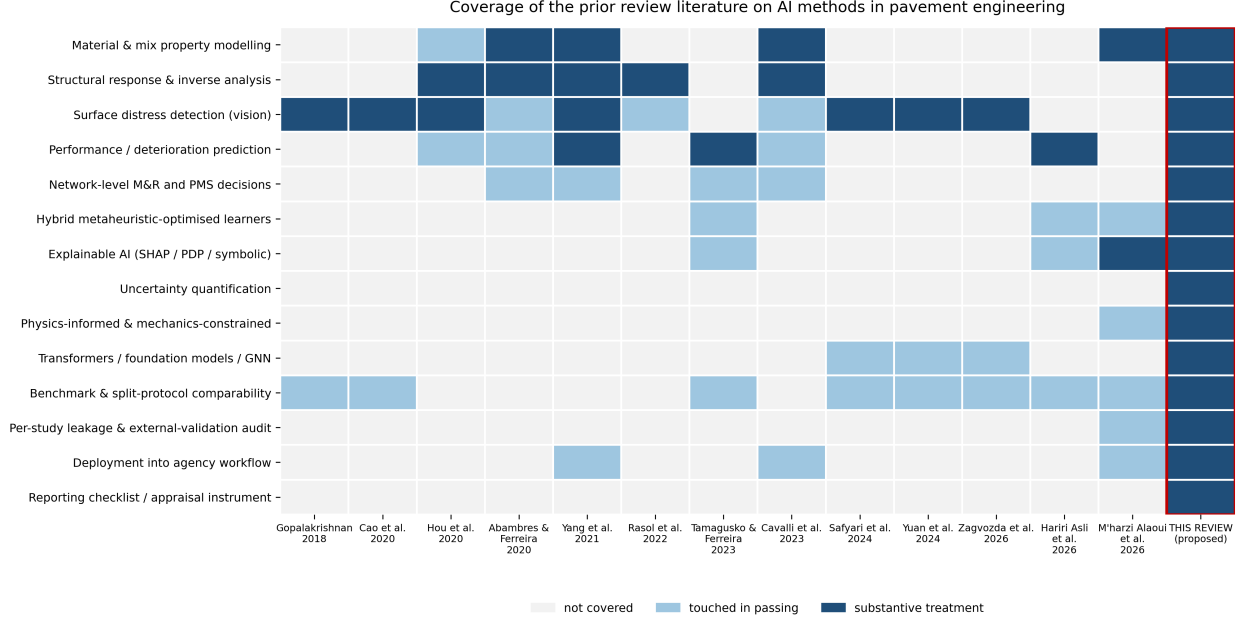


Figure 1: Coverage of the prior review literature on artificial-intelligence methods in pavement engineering. Scoring: 2 = substantive dedicated treatment, 1 = touched in passing, 0 = absent. “This review” is scored on stated scope, to be verified against the completed manuscript at submission.

1.2. Contributions

This review makes four contributions. First, a **structural rather than lexical** definition of hybridity (§2.3) and an algorithm-driven search protocol built to retrieve studies regardless of how their authors chose to describe them. Second, a **taxonomy of seven hybridisation types** (H1–H7, §4), each with a stated theoretical justification and a characteristic failure mode, replacing the field’s undifferentiated use of “hybrid” as a single label. Third, an empirical estimate of the **hybridisation premium** — the performance gain attributable to hybridisation once the comparison is made against a fairly tuned baseline — reported as a distribution across the coded corpus rather than as a single number, since most studies do not report the comparison required to compute it at all (§9). Fourth, a reporting and appraisal instrument, **PAVE-ML**, extending a general data-driven-pavement-model checklist with four items specific to hybrid pipelines (§10), so that the next study in this field can report what this review finds missing from most of the last two decades of it.

2. Review methodology

Full draft in docs/02_MANUSCRIPT_S2_methodology.md; transferred here section by section as each is finalised.

2.1. Scope and framing question

This review asks how much of the performance gain attributed to hybridisation in the pavement literature survives a fair comparison against a properly tuned base learner. The question differs from “where has hybridisation been applied” — the question earlier surveys answer — in that it requires reading each study for a specific comparison rather than for topical coverage, and it requires a definition of hybridity that does not depend on the authors’ own labelling (§2.3).

2.2. Information sources and search

Records were identified from OpenAlex, Scopus, Web of Science Core Collection and Semantic Scholar, with backward and forward snowballing from the 14 prior reviews listed in Table 1. The window runs from 1 January 2005 to 30 June 2026.

Table 1: Prior reviews of artificial-intelligence methods in pavement engineering, with the territory each occupies.

Review	Year	Venue	Territory
Gopalakrishnan	2018	Data	distress-detection
Cao et al.	2020	IEEE Access	distress-detection
Hou et al.	2020	Engineering	structural-evaluation/distress-detection
Abambres et al.	2020	TechRxiv (preprint)	all
Yang et al.	2021	J. Traffic Transp. Eng. (Engl. Ed.)	all
Rasol et al.	2022	Constr. Build. Mater.	structural-evaluation
Cavalli et al.	2023	J. Road Eng.	all
Tamagusko et al.	2023	Infrastructures	performance-prediction
Deng et al.	2023	J. Infrastruct. Preserv. Resil.	performance-prediction
Yuan et al.	2024	Remote Sensing	distress-detection
Safyari et al.	2024	Sensors	distress-detection

Table 1: Prior reviews of artificial-intelligence methods in pavement engineering, with the territory each occupies.

Review	Year	Venue	Territory
M’harzi Alaoui et al.	2026	J. Umm Al-Qura Univ. Eng. Archit.	materials
Zagvozda et al.	2026	Adv. Civ. Archit. Eng.	distress-detection
Hariri Asli et al.	2026	J. Infrastruct. Preserv. Resil.	performance-prediction

2.3. Eligibility

Structural definition of hybridity. A study is in scope if its predictive pipeline couples two or more components drawn from different methodological families, at least one of which is a learned data-driven model, where the coupling is claimed or demonstrated to affect predictive performance. Three operational clarifications resolve most borderline cases. Hyperparameter search counts as hybridisation only when the search procedure is itself a metaheuristic or a surrogate-based optimiser (e.g. particle swarm optimisation, a genetic algorithm, Bayesian optimisation, a tree-structured Parzen estimator); manual tuning and exhaustive grid search do not qualify a study as hybrid, though both are recorded, since they are the baseline against which the premium (§9) is measured. An ensemble of models from a single family (e.g. a random forest) is not a hybrid; a stacked ensemble with heterogeneous base learners is (type H5, §4). Self-description is irrelevant in both directions: a study is included or excluded on the structure of its pipeline, never on whether the authors used the word “hybrid”.

This structural test was necessary, not a refinement for its own sake. A pilot search restricted to hybrid-labelled vocabulary failed to retrieve at least one widely cited, structurally qualifying study — a dipper-throated-optimisation random forest for pothole classification [8] — while an algorithm-name-driven search retrieved it immediately. The same algorithm-name search, however, drifted heavily into adjacent domains (geopolymer concrete, rock-burst prediction, blasting-induced flyrock, and battery remaining-useful-life studies all appeared for generic swarm-optimisation queries), which is why every query in the harvesting protocol (`analysis/harvest_openalex.py`) carries a mandatory, hard-filtered pavement-domain term rather than relying on the optimisation vocabulary alone.

Peer review and language. Peer-reviewed journal articles and full conference papers in English are eligible. Preprints are excluded unless a peer-reviewed version is traced; where cited as grey literature (e.g. [9]) that status is stated explicitly.

Exclusions that shaped the corpus. Three exclusions recur often enough to state here rather than only

in the PRISMA log. Non-neural, non-metaheuristic machine learning without any coupled component — a single tuned random forest or gradient-boosted tree, for instance — is not itself in scope, but is retained in a context set where it serves as a same-paper tuned-baseline comparator (this is, in fact, where several of the review’s most useful premium comparisons come from — see [1] and [10]). Studies where a metaheuristic and a learner are each named in the title but run as separate, non-interacting competing models rather than coupled into one pipeline are excluded from the hybrid corpus proper under the structural test above, and are flagged as an instance of the vocabulary trap in reverse — the label suggests a hybrid that the method does not deliver (e.g. [11], titled around “genetic algorithm and artificial neural networks” but reporting the two as independently competing models). Reinforcement-learning formulations for network-level maintenance and rehabilitation decisions (e.g. [12]) coordinate a learned policy with a simulated or predicted environment in a way that does not map cleanly onto the H1–H7 taxonomy, which is built around single-prediction pipelines; these are retained as context for §9 and §11 rather than coded into the taxonomy.

3. Bibliometric landscape

Of the 56 primary studies coded so far, Figure 2 shows their distribution across the seven hybridisation types defined in §4. A substantial share — 22 of 56 — do not qualify as hybrid under the structural test in §2.3 despite frequent adjacency to hybrid work in citation networks and, in a few cases, despite carrying “hybrid” or an optimiser name in the title while running the two components as separate, non-interacting models (§2.3). This is an early indication, not yet a corpus-wide estimate, of how much daylight exists between the field’s vocabulary and its actual methodological content; §9 returns to the consequence of that gap for the premium estimate.

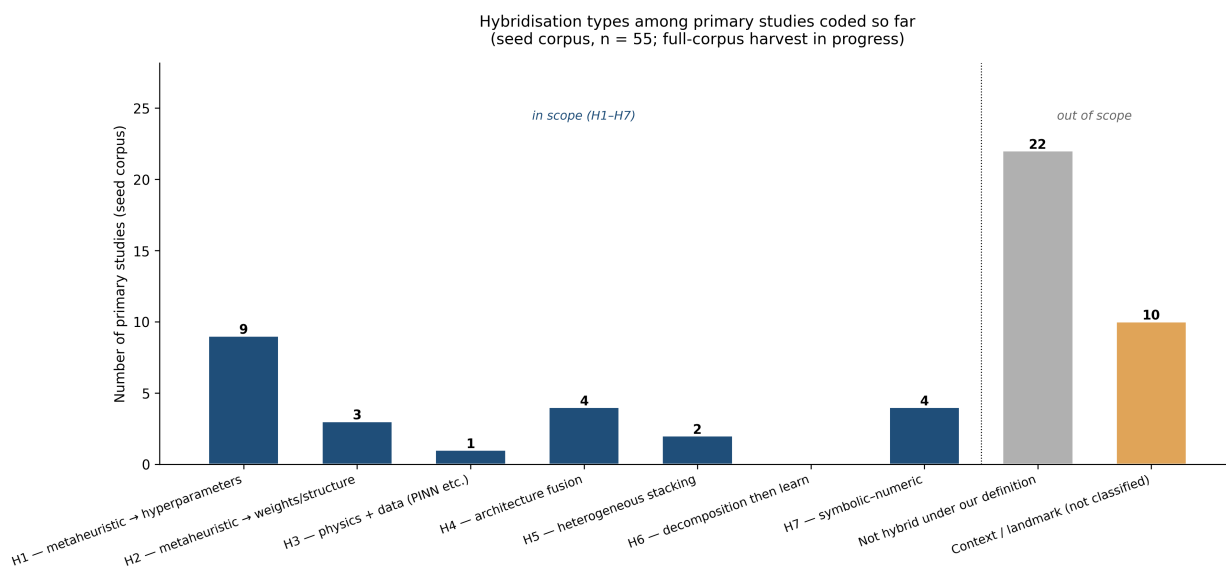


Figure 2: Distribution of hybridisation types among primary studies coded in the seed corpus.

4. Taxonomy of hybridisation

The field’s literature treats “hybrid” as a single, undifferentiated label. This flattens together couplings with different theoretical justifications and, correspondingly, different characteristic failure modes — a metaheuristic searching a hyperparameter space is a different claim from a metaheuristic searching a weight space, and a different claim again from a physics-constrained loss function. Table 2 sets out the seven types this review distinguishes and codes against.

Table 2: Taxonomy of hybridisation in data-driven pavement models (H1–H7).

Type	Coupling	Theoretical justification	Characteristic failure mode
H1	Metaheuristic $\rightarrow$ hyperparameters	Searches a non-convex, discrete hyperparameter space that gradient methods cannot address	Compared against an untuned base learner, so the reported gain is the tuning, not the metaheuristic
H2	Metaheuristic $\rightarrow$ weights / structure	Escapes local minima of backpropagation; permits non-differentiable objectives	Vastly more function evaluations than the baseline receives, with no compute-parity reporting
H3	Physics + data (PINN, mechanistic residual, transfer function as loss)	Constrains the solution to a mechanically admissible manifold; aids under data scarcity	Physics-term weight tuned on the test set; constraint too weak to bind in practice
H4	Architecture fusion (e.g. CNN+transformer, two-stream)	Complementary inductive biases – local texture and long-range context together	Gains confounded with parameter count and training budget rather than the fusion itself
H5	Heterogeneous stacking / blending	Decorrelated errors across model families reduce variance	Meta-learner trained on in-fold predictions; leakage through the stacking layer

Table 2: Taxonomy of hybridisation in data-driven pavement models (H1–H7).

Type	Coupling	Theoretical justification	Characteristic failure mode
H6	Signal decomposition then learn (VMD, EMD, CEEMDAN, wavelet)	Separates scales the learner would otherwise have to disentangle unaided	Decomposition fitted on the full series before the temporal split – the most common severe leakage in this family
H7	Symbolic-numeric (GEP, MGGP, EPR + a numeric optimiser)	Yields a closed-form, inspectable expression rather than an opaque model	Expression complexity uncontrolled; an overfit equation is then read as if it were physics

Two coding notes follow directly from applying this taxonomy to the seed corpus. First, a single study can legitimately receive more than one code — [13] fits both a GA-derived symbolic equation (H7-adjacent) and an ANN whose weights are set by the same GA (H2) on the same data, which is precisely the kind of within-paper comparison §9 is built to exploit. Second, several studies that use metaheuristic or architecture-fusion vocabulary in their titles fail the structural test on inspection — [11] names both a genetic algorithm and an artificial neural network but reports them as independently competing models rather than a coupled pipeline, and is coded **none** rather than any of H1–H7 as a result. Both notes matter for the same reason: the taxonomy is applied to what a pipeline *does*, not to what its title or abstract *calls* it.

Reinforcement-learning approaches to network-level maintenance-and-rehabilitation policy [12, 9] sit adjacent to this taxonomy but are not coded into it: they couple a learned policy to a simulated or predicted environment in a way that does not map onto the single-prediction-pipeline structure H1–H7 assumes. They are retained as context for the decision-consequence discussion in §11 rather than folded into the hybridisation-type counts in §3.

5. Hybrid models in material and mix modelling

PLACEHOLDER.

6. Hybrid models in structural evaluation and inverse analysis

PLACEHOLDER — anchored on back-calculation [14, 15, 16].

7. Hybrid models in performance prediction

PLACEHOLDER.

8. Physics-informed, fusion, stacking, decomposition and symbolic hybrids

PLACEHOLDER — [17] for the physics-informed case; [18] for the symbolic–numeric case.

9. The hybridisation premium

9.1. Definition

For a study reporting a hybrid model M_{hyb} (optimiser $\oplus$ base learner M_{base}) evaluated on a held-out partition, the **hybridisation premium** is

$$\Delta = \text{performance}(M_{\text{hyb}}) - \text{performance}(M_{\text{base}}^{\text{tuned}})$$

where $M_{\text{base}}^{\text{tuned}}$ is the same base learner, tuned by a conventional procedure under a **comparable evaluation budget**, evaluated on the **same** partition with the **same** metric. All three constraints are load-bearing: a premium computed against an untuned, default-setting baseline is not a measure of what hybridisation contributes — it is a measure of what tuning contributes, mislabelled.

9.2. What the completed audit will report

Over the full coded corpus, the review will report, for every study attempting an H1 or H2 coupling: whether Δ is computable at all under the definition above (**premium_computable**); its value where it is; whether the search budget was stated for both the hybrid and the baseline arm (**budget_parity_reported**); and, where a newly named optimiser is the paper’s contribution, whether it was benchmarked against at least two established optimisers on the same problem (**optimiser_novelty_claim**). These fields are specified in full, with worked examples, in the PAVE-ML instrument (§10) and were fixed before full-text coding began.

9.3. Preliminary evidence from the seed corpus

The completed audit is not yet available; what follows is the evidence already visible in the seed corpus, presented as motivation for the audit rather than as its result. Table 3 collects every within-paper comparison in the seed corpus close enough to a premium calculation to be informative, with every figure copied verbatim from the source paper’s own reported results.

Table 3: Within-corpus evidence bearing on the hybridisation premium. Every figure is copied verbatim from the cited paper’s own reported results; none is computed by the present review.

Source	Comparison	Metric	Non-hybrid		Premium	Note
			value	Hybrid value		
10.1038/s41598-022-17429-z	1.5S2M17429-zR2 / RMSE (best single-kernel setting reported) vs. best of 6 swarm-optimised LSSVM variants	(MPa)	not reported as a standalone tuned baseline	0.942 / 6.72 (best of six, SOS)	not computable — no tuned non-hybrid baseline reported	The paper compares six hybrids to each other, never to a conventionally tuned LSSVM. This is the missing-baseline pattern Section 9 quantifies.
10.28991/cej-2025-011-01-062	2025-011-01-062 (dynamic individually-tuned learners (ANN, RNN, CNN among them) vs. Witczak/Hirsch mechanistic baselines)	R2 (dynamic modulus)	Witczak 1-40D / Hirsch (mechanistic)	bagging ensemble, individually tuned (highest of eight)	positive but modest — mechanistic models remain competitive; deep architectures do NOT win	The field’s clearest example of tuning parity: every learner gets its own search budget, and the winner is a tuned ensemble tree, not a hybrid or a deep net.

Table 3: Within-corpus evidence bearing on the hybridisation premium. Every figure is copied verbatim from the cited paper’s own reported results; none is computed by the present review.

Source	Comparison	Metric	Non-hybrid		Premium	Note
			value	Hybrid value		
10.1016/j.sand.2020.102010	GA-2020 symbolic equation (H7- adjacent) vs. ANN-GA (GA on ANN weights, H2)	R2 (resilient modulus)	GA-only symbolic model	ANN-GA hybrid	ANN-GA reported higher R2, but adds a black-box layer over an already- competitive symbolic model	Rare true head-to- head between two hybrid types on identical data — flagged for full-text extraction of exact R2 values.
10.3390/app916321	PSO-ANN vs. PSO- ELM vs. kernel- ELM, all reported within one paper	RMSE / R2	PSO-ANN (same optimiser, different base learner)	PSO-ELM (best reported)	small, same- optimiser architecture comparison — isolates base-learner choice, not optimiser value	Useful for a different question than the premium: it holds the optimiser constant and varies the learner, the mirror image of what Section 9 needs.

Table 3: Within-corpus evidence bearing on the hybridisation premium. Every figure is copied verbatim from the cited paper’s own reported results; none is computed by the present review.

Source	Comparison	Metric	Non-hybrid		Premium	Note
			value	Hybrid value		
10.3390/ma181209418	ML models (KNN, Bayesian ridge, decision tree) vs. stacking ensemble	R2 (G* and phase angle)	best single model	stacking (R2 = 0.973 / 0.999)	positive, and the ONLY row in this table where leakage is explicitly controlled by design	Cross-validated meta-features stated as a deliberate leakage-avoidance choice — the paper we cite as the positive PAVE-ML exemplar.

Two readings of this table matter more than any single row. First, **the comparison the premium requires is usually absent, not unfavourable**. Of the five rows, only two [2, 19] involve a paper that reports enough to attempt a premium calculation at all, and [2]’s own six-optimiser comparison never includes the one comparator — a conventionally tuned, non-hybrid version of the same LSSVM — that would let a reader compute Δ . This is the pattern the full audit is built to quantify: the missing baseline, not a negative result, is the dominant finding so far. Second, **where tuning parity is closest to being honoured, the hybrid or deep-architecture advantage narrows**. [1] tunes eight architectures individually, including deep networks, against both each other and two established mechanistic models; the winner is a tuned bagging ensemble. Neither a hybrid nor a deep architecture prevailed once every competitor received its own search budget. [19] is, so far, the seed corpus’s clearest positive counter-example — a heterogeneous stacking ensemble (H5) that both reports a credible premium over single models and states explicitly that its meta-learner was trained on cross-validated, not in-fold, base-model predictions specifically to avoid the leakage pathway characteristic of that hybridisation type (§4, H5). It is cited in §10 as the positive PAVE-ML exemplar for exactly this reason.

Neither reading should be generalised past what five studies can support. They are reported here as the

reason the systematic audit in this section is worth completing, not as its substitute.

10. Interpretability and uncertainty in hybrid pipelines

PLACEHOLDER.

11. From premium to decision

A premium, once measured, still needs to be worth having. Hosseini and Smadi [20] provide the sharpest available answer to that question, though the paper itself is about prediction error rather than hybridisation as such: injecting 10–90% error into a pavement performance prediction and propagating it through a 20-year network-level maintenance and rehabilitation schedule, they price the resulting treatment-cost consequence directly. That result supplies the yardstick a hybridisation premium should be measured against — not whether $\Delta > 0$, but whether Δ is large enough, at the accuracy level the underlying prediction task already operates at, to change which treatment gets scheduled, or when. A premium of a few hundredths of a unit of R^2 , of the kind several rows in Table 3 report or gesture toward, is a strong candidate for being decision-irrelevant by this standard; the completed audit will report, for every study where both a premium and enough contextual detail are available, whether the premium plausibly clears this bar or not. This section will be completed once that judgement can be made study by study rather than asserted in general.

12. PAVE-ML: a reporting and appraisal checklist

The audit this review conducts is only as trustworthy as the coding rules behind it, and those rules are only useful to the field if they survive as something other than this paper’s internal methodology. PAVE-ML is offered as both: the instrument this review scores every included study against, and a checklist future studies can score themselves against before submission.

The full instrument — twenty items across five domains (data provenance, preprocessing and partitioning, modelling and comparison, evaluation and interpretation, and generalisation and transparency), plus four items specific to hybrid pipelines — is reproduced in full in the supplementary material (`docs/03_PAVE-ML_instrument.md`), together with the operational decision rules and worked examples that make the coding reproducible rather than a matter of reader judgement. Two design choices are worth stating here because they determine how the audit’s headline numbers should be read. First, **silence is coded as absence, not as unclear** — if a study does not state that its test partition was held out independently of hyperparameter selection, that is coded **no**, following the convention used in clinical prediction-model appraisal (TRIPOD, PROBAST), because it is the only convention that cannot be accused of charitable reading. Second, the four hybrid-specific items require a study to report its base learner’s

performance both with and without the hybridising component on the same partition and metric (item 12a); to state the search budget for both arms so that parity can be assessed (12b); to run stochastic optimisers more than once and report the spread rather than the best run (12c); and, where a newly named optimiser is the contribution, to benchmark it against at least two established optimisers on the same problem (12d). [19] is, at the time of writing, the clearest example in the seed corpus of a study satisfying the spirit of items in this list without having been written against it — its stacking meta-learner is trained on cross-validated, not in-fold, predictions specifically to avoid leakage, which is exactly the practice item 7 of the general instrument (grouped/blocked partitioning respected by all fitted preprocessing) is written to detect the absence of.

The reliability of the instrument itself will be reported, not assumed: a random 15% of the fully coded corpus will be independently double-coded, Cohen’s κ reported per field, and any field falling below $\kappa = 0.60$ will have its decision rule rewritten and be recoded in full rather than reported with a caveat. This procedure, and its result, are reported in full in the completed manuscript.

13. Research agenda

PLACEHOLDER.

14. Conclusions

PLACEHOLDER.

Declaration of competing interest

The authors declare no competing financial interests or personal relationships that could have influenced the work reported in this paper.

Declaration of generative AI in the writing process

PLACEHOLDER — to be completed per the target journal’s wording before submission. Elsevier requires a statement naming the tool, the purpose of use, and confirmation that the authors reviewed and take full responsibility for the content. Disclosed use is permitted; undisclosed use is not.

Data availability

The screening database, coding rubric, harvesting scripts and the full analysis pipeline that produced every figure and table in this paper are available at [REPOSITORY URL].

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